Technical Note: block-DDA_Py

Discrete Dipole Approximation with FFT-accelerated
Block-Krylov Solver for Light Scattering by Arbitrary Particles

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1 Introduction

The discrete dipole approximation (DDA) is a flexible numerical method for computing electromagnetic scattering and absorption by particles of arbitrary shape and composition [1, 2]. The target particle is represented by an array of N point dipoles on a cubic lattice with spacing d. Each dipole acquires a polarisation in response to the local electric field, which consists of the incident wave plus the fields radiated by all other dipoles.

This document describes the theoretical background and the algorithms that are implemented in version 0.3.0 of `block-DDA_Py`, covering:

1. the DDA formulation and polarisability prescription (Sec. 2),
2. the Goodman (1991) FFT-accelerated matrix–vector product (Sec. 3),
3. the block-Krylov iterative solvers (Sec. 4),
4. the scattering observables computed from the solution (Sec. 5),
5. the Gaussian Random Ellipsoid (GRE) shape model (Sec. 6),
6. the Mie theory reference solution for homogeneous spheres (Sec. 7).

Throughout this note, the electromagnetic equations are written in Gaussian (CGS) units; lengths are expressed in micrometers (µm). Vectors are set in bold italic ($\boldsymbol{E}$, $\boldsymbol{k}$, $\boldsymbol{r}$), matrices and second-rank tensors in upright bold ($\mathbf{A}$, $\boldsymbol{\alpha}$), and unit vectors carry a hat ($\hat{\boldsymbol{e}}$, $\hat{\boldsymbol{r}}$).

2 DDA Formulation

2.1 Fundamental equation

Consider a monochromatic plane wave of vacuum wavelength λ_0 incident on a particle embedded in a non-absorbing medium of refractive index m_m. The wavenumber in the medium is

$$k = \frac{2\pi\, m_m}{\lambda_0}. \tag{1}$$

The particle is discretised into N cubic elements (dipoles) of side length d and volume $v = d^3$, placed on a regular lattice with grid dimensions $N_x \times N_y \times N_z$. The lattice position of the i-th dipole is $\boldsymbol{r}_i$.

The polarisation $\boldsymbol{P}_i$ of dipole i satisfies

$$\boldsymbol{P}_i = \boldsymbol{\alpha}_i \cdot \boldsymbol{E}(\boldsymbol{r}_i), \tag{2}$$

where $\boldsymbol{\alpha}_i$ is the 3×3 polarisability tensor and $\boldsymbol{E}(\boldsymbol{r}_i)$ is the total electric field at position $\boldsymbol{r}_i$. The total field is the superposition of the incident field and the contributions from all other dipoles:

$$\boldsymbol{E}(\boldsymbol{r}_i) = \boldsymbol{E}_{\mathrm{inc},i} - \sum_{j \neq i} \mathbf{G}(\boldsymbol{r}_i - \boldsymbol{r}_j) \cdot \boldsymbol{P}_j, \tag{3}$$

where $\mathbf{G}$ is the free-space dyadic Green's function (Sec. 2.3). Substituting (3) into (2) and rearranging yields the $3N \times 3N$ linear system

$$\mathbf{A}\,\boldsymbol{P} = \boldsymbol{E}_{\mathrm{inc}}, \tag{4}$$

where $\boldsymbol{P}$ and $\boldsymbol{E}_{\mathrm{inc}}$ are column vectors of length $3N$ formed by concatenating the three Cartesian components of each dipole (ordering: $P_{x,0}, P_{y,0}, P_{z,0}, P_{x,1}, \ldots$), and $\mathbf{A}$ is a complex-valued matrix with

$$\mathbf{A}_{ij} = \begin{cases} \boldsymbol{\alpha}_i^{-1} & \text{if } i = j, \\ \mathbf{G}(\boldsymbol{r}_i - \boldsymbol{r}_j) & \text{if } i \neq j. \end{cases} \tag{5}$$

When the dipoles are placed on a regular cubic lattice, the off-diagonal blocks $\mathbf{A}_{ij}$ depend only on the difference $\boldsymbol{r}_i - \boldsymbol{r}_j$, giving $\mathbf{A}$ a block-Toeplitz structure that can be exploited by FFT (Sec. 3).

2.2 Incident field

The incident plane wave in the medium is

$$\boldsymbol{E}_{\mathrm{inc},i} = \boldsymbol{E}_0 \exp(i\boldsymbol{k} \cdot \boldsymbol{r}_i - i\omega t), \tag{6}$$

where $\boldsymbol{k} = k\,\hat{\boldsymbol{u}}_{\mathrm{inc}}$ with $\hat{\boldsymbol{u}}_{\mathrm{inc}}$ the propagation direction (fixed to $+z$ in the laboratory frame). The time-harmonic factor $e^{-i\omega t}$ is suppressed hereafter.

To simulate orientation-averaged scattering, the particle is rotated through L orientations parametrised by intrinsic ZYZ Euler angles $(\alpha_\ell, \beta_\ell, \gamma_\ell)$, $\ell = 1, \ldots, L$. For each orientation, the laboratory-frame vectors are transformed into the particle frame via the inverse rotation matrix. Batch rotation is performed using `scipy.spatial.transform.Rotation`.

Following the CAS-v2 (Complex Amplitude Sensing, version 2) beam configuration, the incident polarisation is assumed to be left-handed circular. This setting is hard-coded in the current version (v0.3.0). The polarisation vector is:

$$\hat{\boldsymbol{e}}_0 = \frac{1}{\sqrt{2}}\,\hat{\boldsymbol{e}}_\theta + \frac{i}{\sqrt{2}}\,\hat{\boldsymbol{e}}_\phi, \tag{7}$$

where $\hat{\boldsymbol{e}}_\theta$ and $\hat{\boldsymbol{e}}_\phi$ are the polar and azimuthal unit vectors associated with the incident direction.

2.3 Free-space dyadic Green's function

The 3×3 interaction tensor for two dipoles separated by $\boldsymbol{r} = \boldsymbol{r}_i - \boldsymbol{r}_j$ ($r = |\boldsymbol{r}|$, $\hat{\boldsymbol{r}} = \boldsymbol{r}/r$) is [3]

$$\mathbf{G}(\boldsymbol{r}) = \frac{e^{ikr}}{r}\left[k^2\big(\hat{\boldsymbol{r}} \otimes \hat{\boldsymbol{r}} - \mathbf{I}_3\big) + \frac{1 - ikr}{r^2}\big(\mathbf{I}_3 - 3\,\hat{\boldsymbol{r}} \otimes \hat{\boldsymbol{r}}\big)\right], \tag{8}$$

where $\otimes$ denotes the outer (tensor) product and $\mathbf{I}_3$ the 3×3 identity matrix. The self-interaction ($\boldsymbol{r} = \mathbf{0}$) is excluded from $\mathbf{G}$ and handled separately via the diagonal $\boldsymbol{\alpha}_i^{-1}$ in (5).

2.4 Polarisability: Clausius–Mossotti with radiative correction

The relative permittivity of the i-th dipole along Cartesian axis c ($c \in \{x, y, z\}$) is

$$\epsilon_{r,c} = \frac{m_{p,c}^2}{m_m^2}, \tag{9}$$

where $m_{p,c}$ is the (generally complex) particle refractive index along axis c.

The static Clausius–Mossotti (CM) polarisability is

$$\alpha_{0,c} = \frac{3}{4\pi}\,\frac{\epsilon_{r,c} - 1}{\epsilon_{r,c} + 2}\,v, \tag{10}$$

where $v = d^3$ is the dipole volume.

To improve accuracy, the CM polarisability is corrected for radiative reaction following Chaumet & Rahmani [4]. Define the equivalent-volume sphere radius

$$a = \left(\frac{3v}{4\pi}\right)^{1/3} \tag{11}$$

and the correction term

$$M = \frac{8\pi}{3}\big[(1 - ika)\, e^{ika} - 1\big]. \tag{12}$$

The dynamical (radiative-reaction corrected) polarisability is then

$$\alpha_c = \frac{\alpha_{0,c}}{1 - M\,\alpha_{0,c}\,/\,v}. \tag{13}$$

In general the particle may be optically anisotropic, so $\boldsymbol{\alpha}_i = \mathrm{diag}(\alpha_x, \alpha_y, \alpha_z)$. For isotropic particles, $m_{p,x} = m_{p,y} = m_{p,z}$ and the polarisability reduces to a scalar times $\mathbf{I}_3$.

3 FFT-Accelerated Matrix–Vector Product (Goodman 1991)

Direct evaluation of the matrix–vector product (MVP) $\mathbf{A}\,\boldsymbol{P}$ requires $O(N^2)$ operations and $O(N^2)$ storage. Because the off-diagonal part of $\mathbf{A}$ has block-Toeplitz structure on a regular cubic lattice, the MVP can be evaluated via 3D FFT in $O(N \log N)$ time with $O(N)$ storage [3].

3.1 Interaction tensor on the doubled grid

The physical cuboid lattice has dimensions $N_x \times N_y \times N_z$. The index difference along each axis α ranges from $-(N_\alpha - 1)$ to $+(N_\alpha - 1)$. To evaluate the linear convolution as a cyclic convolution via FFT, the lattice is extended to $2N_x \times 2N_y \times 2N_z$ with the following index mapping.

For each extended-grid index $I_\alpha \in \{0, 1, \ldots, 2N_\alpha - 1\}$, the physical integer displacement $\Delta\alpha$ is

$$\Delta\alpha(I_\alpha) = \begin{cases} I_\alpha & 0 \le I_\alpha \le N_\alpha - 1 \quad \text{(positive displacements)}, \\ 0 & I_\alpha = N_\alpha \quad \text{(Nyquist; zeroed out)}, \\ I_\alpha - 2N_\alpha & N_\alpha + 1 \le I_\alpha \le 2N_\alpha - 1 \quad \text{(negative displacements)}. \end{cases} \tag{14}$$

The physical displacement vector at extended-grid index (I_x, I_y, I_z) is $\boldsymbol{r} = d\,(\Delta x,\, \Delta y,\, \Delta z)$, and the interaction tensor $\mathbf{G}(\boldsymbol{r})$ is computed via (8). Three special cases are set to zero:

- self-interaction at the origin: $\mathbf{G}(\mathbf{0}) = \mathbf{0}$,
- Nyquist positions ($I_\alpha = N_\alpha$ for any α): physically unreachable displacements.

The resulting array has shape $(2N_x, 2N_y, 2N_z, 3, 3)$ and dtype `complex128`.

3.2 Pre-computation of the FFT

Before the iterative solver begins, the 3D discrete Fourier transform of $\mathbf{G}$ along the spatial axes is computed once:

$$\widetilde{\mathbf{G}} = \mathrm{FFT}_{3\mathrm{D}}\big[\mathbf{G}\big]. \tag{15}$$

$\widetilde{\mathbf{G}}$ has the same shape $(2N_x, 2N_y, 2N_z, 3, 3)$ and is stored in memory throughout the simulation. It is independent of the incident direction, polarisation, and Krylov iteration, so this computation is performed only once per (λ_0, m_m) pair.

3.3 Block MVP via FFT convolution

Given a block of L polarisation vectors stacked as columns in $\mathbf{P}_{\mathrm{block}}$ (shape $3N_xN_yN_z \times L$), the off-diagonal MVP $\mathbf{A}_{\mathrm{off}}\,\mathbf{P}_{\mathrm{block}}$ is computed as follows.

1. **Reshape** $\mathbf{P}_{\mathrm{block}}$ from $(3N_xN_yN_z,\, L)$ to the 5D physical representation $(N_x, N_y, N_z, 3, L)$.

2. **Zero-pad** to the doubled grid:

$$\hat{\mathbf{P}}(I_x, I_y, I_z, c, \ell) = \begin{cases} \mathbf{P}(I_x, I_y, I_z, c, \ell) & \text{if } I_\alpha < N_\alpha \ \forall \alpha, \\ 0 & \text{otherwise.} \end{cases} \tag{16}$$

3. **Forward 3D FFT** along spatial axes: $\widetilde{\mathbf{P}} = \mathrm{FFT}_{3\mathrm{D}}[\hat{\mathbf{P}}]$.

4. **Pointwise tensor product** in Fourier space:

$$\widetilde{Y}_{i,\ell} = \sum_{j=1}^{3} \widetilde{G}_{ij} \, \widetilde{P}_{j,\ell} \qquad \text{(at each spatial frequency)}, \tag{17}$$

   i.e. a 3×3 matrix times a $3 \times L$ matrix at every grid point.

5. **Inverse 3D FFT**: $\hat{\mathbf{Y}} = \mathrm{FFT}_{3\mathrm{D}}^{-1}[\widetilde{\mathbf{Y}}]$.

6. **Extract** the physical region $\hat{\mathbf{Y}}[{:}N_x, {:}N_y, {:}N_z, :, :]$ and flatten back to $(3N_xN_yN_z, L)$.

The full preconditioned MVP including the diagonal is

$$\mathbf{A}\,\boldsymbol{S} = \mathrm{diag}(\mathbf{A}) \cdot \boldsymbol{S} + \mathbf{A}_{\mathrm{off}}\,\boldsymbol{S}, \tag{18}$$

where $\mathrm{diag}(\mathbf{A}) = \boldsymbol{\alpha}^{-1}$ is applied element-wise and $\mathbf{A}_{\mathrm{off}}$ is evaluated via the FFT procedure above.

3.4 Computational complexity and memory

- **Time**: Each MVP costs $O(N_{\mathrm{cuboid}} \log N_{\mathrm{cuboid}})$ where $N_{\mathrm{cuboid}} = N_xN_yN_z$. For the block variant with L right-hand sides, the L columns are processed in a single batched FFT call.
- **Memory**: The dominant allocations are:
  - $\widetilde{\mathbf{G}}$: $2N_x \times 2N_y \times 2N_z \times 9$ complex128 entries $= 1152\,N_{\mathrm{cuboid}}$ bytes (static, allocated once);
  - padded polarisation arrays (forward and inverse FFT): $2N_x \times 2N_y \times 2N_z \times 3 \times L$ complex128 entries each $= 384\,L\,N_{\mathrm{cuboid}}$ bytes (two such arrays at peak).

  Peak memory $\approx (1152 + 768\,L)\,N_{\mathrm{cuboid}}$ bytes.

3.5 Multi-threaded FFT

The 3D FFTs are computed by `scipy.fft.fftn` with multi-threading. The default number of worker threads is $\max(1,\ n_{\mathrm{cpu}} - 2)$, leaving two cores free for other processes. This can be overridden at runtime via the environment variable `DDA_FFT_WORKERS`.

4 Block-Krylov Iterative Solvers

The DDA system (4) is solved for L right-hand sides simultaneously. Stacking the L incident-field vectors as columns of a block matrix $\mathbf{B}$ (shape $3N \times L$), the problem becomes

$$\mathbf{A}\,\mathbf{X} = \mathbf{B}, \tag{19}$$

where $\mathbf{X}$ (shape $3N \times L$) contains the L solution vectors.

4.1 Jacobi preconditioning

A left Jacobi (diagonal) preconditioner is applied:

$$\mathbf{M}^{-1} = \mathrm{diag}(\mathbf{A})^{-1} = \mathrm{diag}(\alpha_{c,i}), \tag{20}$$

so that the preconditioned system becomes $\mathbf{M}^{-1}\mathbf{A}\,\mathbf{X} = \mathbf{M}^{-1}\mathbf{B}$.

4.2 Block-BiCGSTAB

The primary solver is Block-BiCGSTAB [5], which extends the stabilised bi-conjugate gradient method to block systems. It is suitable for general (non-symmetric) matrices and is the default in `block-DDA_Py`.

The algorithm is summarised in Table 1. Each iteration requires two block MVPs (lines 1 and 4). The default convergence parameters are $\tau = 0.01$ and $n_{\max} = 25$.

Table 1: Block-BiCGSTAB with Jacobi preconditioning (after Tadano et al. [5]).

Input: $\mathbf{A}$, $\mathbf{B}$ (shape $3N \times L$), $\mathbf{M} = \mathrm{diag}(\mathbf{A})$, tolerance τ, max iterations $n_{\max}$.

Initialise: $\mathbf{X}_0 = \mathbf{0}$, $\mathbf{R}_0 = \mathbf{M}^{-1}\mathbf{B}$, $\mathbf{P}_0 = \mathbf{R}_0$, $\widetilde{\mathbf{R}}_0 = \mathbf{R}_0$.

For $n = 0, 1, \ldots, n_{\max} - 1$:

1. $\mathbf{V}_n = \mathbf{M}^{-1}\mathbf{A}\,\mathbf{P}_n$
2. $\boldsymbol{\alpha}_n = (\widetilde{\mathbf{R}}_0^{\mathsf{H}}\,\mathbf{V}_n)^{-1}\,\widetilde{\mathbf{R}}_0^{\mathsf{H}}\,\mathbf{R}_n$
3. $\mathbf{T}_n = \mathbf{R}_n - \mathbf{V}_n\,\boldsymbol{\alpha}_n$
4. $\mathbf{Z}_n = \mathbf{M}^{-1}\mathbf{A}\,\mathbf{T}_n$
5. $\xi_n = \mathrm{tr}(\mathbf{Z}_n^{\mathsf{H}}\,\mathbf{T}_n)\ /\ \mathrm{tr}(\mathbf{Z}_n^{\mathsf{H}}\,\mathbf{Z}_n)$
6. $\mathbf{X}_{n+1} = \mathbf{X}_n + \mathbf{P}_n\,\boldsymbol{\alpha}_n + \xi_n\,\mathbf{T}_n$
7. $\mathbf{R}_{n+1} = \mathbf{T}_n - \xi_n\,\mathbf{Z}_n$
8. If $\|\mathbf{R}_{n+1}\|/\|\mathbf{M}^{-1}\mathbf{B}\| < \tau$: stop, return $\mathbf{X}_{n+1}$.
9. $\boldsymbol{\beta}_n = (\widetilde{\mathbf{R}}_0^{\mathsf{H}}\,\mathbf{V}_n)^{-1}\,(-\widetilde{\mathbf{R}}_0^{\mathsf{H}}\,\mathbf{Z}_n)$
10. $\mathbf{P}_{n+1} = \mathbf{R}_{n+1} + (\mathbf{P}_n - \xi_n\,\mathbf{V}_n)\,\boldsymbol{\beta}_n$

4.3 Block-COCG-rq (alternative solver)

Since the DDA interaction matrix $\mathbf{A}$ is complex symmetric ($\mathbf{A} = \mathbf{A}^{\top}$), a solver specialised for this structure, Block-COCG-rq [6], is also provided. Block-COCG-rq requires only one block MVP per iteration and is therefore faster per iteration than Block-BiCGSTAB. However, for particles with high refractive index it exhibits poorer convergence stability compared to Block-BiCGSTAB. To switch the solver, replace the import and function call in `bl_dda/scatterer.py` (line 3 and line 169): change `bl_bicgstab_jacobi_mvp_fft` to `bl_cocg_rq_jacobi_mvp_fft`. Both functions share the same interface.

5 Scattering Observables

Once the polarisation $\boldsymbol{P}_i$ ($i = 1, \ldots, N$) has been obtained for each orientation ℓ, the following observables are computed. Throughout this section, k denotes the wavenumber in the medium as defined in (1).

5.1 Absorption and extinction cross sections

The absorption cross section is [2]

$$C_{\text{abs}} = 4\pi k \sum_{i=1}^{N} \text{Im}\big(\boldsymbol{P}_i \cdot \boldsymbol{E}_i^*\big), \tag{21}$$

where $\boldsymbol{E}_i$ is the internal electric field at dipole i, reconstructed from the polarisation via

$$\boldsymbol{E}_i = \frac{4\pi\, \boldsymbol{P}_i}{(\epsilon_{r,i} - 1)\, v}. \tag{22}$$

The extinction cross section is

$$C_{\text{ext}} = 4\pi k \sum_{i=1}^{N} \text{Im}\big(\boldsymbol{P}_i \cdot \boldsymbol{E}_{\text{inc},i}^*\big). \tag{23}$$

The corresponding efficiency factors are $Q_{\text{abs}} = C_{\text{abs}}/(\pi r_{ve}^2)$ and $Q_{\text{ext}} = C_{\text{ext}}/(\pi r_{ve}^2)$, where r_{ve} is the volume-equivalent sphere radius:

$$r_{ve} = \left(\frac{3\, N\, v}{4\pi}\right)^{1/3}. \tag{24}$$

5.2 Forward scattering amplitude (PCAS)

The forward scattering amplitude in the PCAS (Polarimetric Complex Amplitude of Scattering) framework is computed from the far-field projection of the dipole polarisations.

For the forward scattering direction $\hat{\boldsymbol{u}}_{\text{fw}} = +\hat{\boldsymbol{z}}$ in the laboratory frame, define the phase factor for each dipole:

$$e_{\text{fw},i} = \exp\big(ik\, \hat{\boldsymbol{u}}_{\text{fw}} \cdot \boldsymbol{r}_i\big). \tag{25}$$

The θ- and ϕ-components of the forward scattering amplitude correspond to the s- and p-polarisation channels of the CAS-v2 detector, respectively: $S_{\text{fw},\theta}^{\text{PCAS}} \equiv S_s(0)$ and $S_{\text{fw},\phi}^{\text{PCAS}} \equiv S_p(0)$. They are computed as

$$S_{\text{fw},\theta}^{\text{PCAS}}\ [= S_s(0)] = k^2 \sum_{i=1}^{N} (\boldsymbol{P}_i \cdot \hat{\boldsymbol{e}}_{\theta,\text{fw}})\ (\sqrt{2}\, e_{\text{fw},i})^*, \tag{26}$$

$$S_{\text{fw},\phi}^{\text{PCAS}}\ [= S_p(0)] = k^2 \sum_{i=1}^{N} (\boldsymbol{P}_i \cdot \hat{\boldsymbol{e}}_{\phi,\text{fw}})\ (\sqrt{2}\, i\, e_{\text{fw},i})^*, \tag{27}$$

where the factors $\sqrt{2}$ and $\sqrt{2}\, i$ arise from the circular-polarisation decomposition of the incident field (7).

5.3 Backward scattering amplitude (OCBS)

For the exact backward scattering direction $\hat{\boldsymbol{u}}_{\text{bk}} = -\hat{\boldsymbol{z}}$, define the phase factor

$$e_{\text{bk},i} = \exp\big(ik\, \hat{\boldsymbol{u}}_{\text{bk}} \cdot \boldsymbol{r}_i\big). \tag{28}$$

Let $\hat{\boldsymbol{e}}_{\theta,\text{bk}}$ and $\hat{\boldsymbol{e}}_{\phi,\text{bk}}$ be the polar and azimuthal unit vectors associated with the backward direction. In the laboratory frame: $\hat{\boldsymbol{e}}_{\theta,\text{bk}} = -\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{e}}_{\phi,\text{bk}} = +\hat{\boldsymbol{y}}$.

The θ- and ϕ-components of the backward scattering amplitude are

$$S_{\mathrm{bk},\theta}^{\mathrm{OCBS}} = k^2 \sum_{i=1}^{N} (\boldsymbol{P}_i \cdot \hat{\boldsymbol{e}}_{\theta,\mathrm{bk}})\, (\sqrt{2}\, e_{\mathrm{bk},i})^*, \tag{29}$$

$$S_{\mathrm{bk},\phi}^{\mathrm{OCBS}} = k^2 \sum_{i=1}^{N} (\boldsymbol{P}_i \cdot \hat{\boldsymbol{e}}_{\phi,\mathrm{bk}})\, (\sqrt{2}\, i\, e_{\mathrm{bk},i})^*, \tag{30}$$

where the reference-wave factors $\sqrt{2}$ and $\sqrt{2}\, i$ are the same as in the forward case.

The OCBS (Opposite-Circular-Basis Scattering) amplitude is then

$$S_{\mathrm{bk}}^{\mathrm{OCBS}} = \frac{-S_{\mathrm{bk},\theta}^{\mathrm{OCBS}} + S_{\mathrm{bk},\phi}^{\mathrm{OCBS}}}{\sqrt{2}}. \tag{31}$$

More precisely, the θ- and ϕ-components correspond to combinations of the Mishchenko amplitude matrix elements [9]:

$$S_{\mathrm{bk},\theta}^{\mathrm{OCBS}} \equiv S_{11}(\pi) + i\, S_{12}(\pi), \qquad S_{\mathrm{bk},\phi}^{\mathrm{OCBS}} \equiv S_{22}(\pi) - i\, S_{21}(\pi). \tag{32}$$

The backscattering theorem $S_{21}(\pi) = -S_{12}(\pi)$ causes the off-diagonal contributions to cancel when the two components are combined in (31), yielding an expression equivalent to Eq. (28) of Moteki and Adachi [10].

6 Gaussian Random Ellipsoid (GRE) Shape Model

The particle shape is generated using the Gaussian Random Ellipsoid (GRE) model of Muinonen & Pieniluoma [7].

6.1 Base ellipsoid

The base ellipsoid is defined by three semi-axes $a \geq b \geq c$ and the volume-equivalent radius r_v. Two axis ratios are specified as input: b/c (`bc_ratio`) and a/b (`ab_ratio`). The semi-axis c is determined by volume conservation:

$$c = \left(\frac{r_v^3}{(a/b)\,(b/c)^2} \right)^{1/3}, \qquad b = c \cdot (b/c), \qquad a = b \cdot (a/b). \tag{33}$$

6.2 Surface deformation

A Gaussian random field $s(\theta, \phi)$ is sampled on the surface of the base ellipsoid with covariance

$$\mathrm{Cov}(s_i, s_j) = \beta^2 \exp\left(-\frac{\|\boldsymbol{r}_i - \boldsymbol{r}_j\|^2}{2\,\ell_c^2} \right), \tag{34}$$

where β is the standard deviation of the deformation and ℓ_c is the correlation length, which controls the fineness of the surface roughness. In the current version (v0.3.0), $\ell_c = 0.3\,c$ is adopted as an empirical choice.

The deformed surface position is obtained by displacing each point on the base ellipsoid along the outward surface normal $\hat{\boldsymbol{n}}_0$:

$$\boldsymbol{r}_{\mathrm{GRE}}(\theta, \phi) = \boldsymbol{r}_{\mathrm{base}}(\theta, \phi) + h_0 \big[\exp\big(s(\theta, \phi)\big) - \tfrac{1}{2}\beta^2 - 1 \big]\, \hat{\boldsymbol{n}}_0(\theta, \phi), \tag{35}$$

where $h_0 = c^2/a$ is a length scale and $\hat{\boldsymbol{n}}_0$ is the unit outward normal to the base ellipsoid.

6.3 Lattice generation and interior detection

A cubic lattice with spacing

$$d = \frac{1}{75}\sqrt{\frac{c}{0.05\,\mu\mathrm{m}}}\,\big[(a/b)\,(b/c)\big]^{1/5} \tag{36}$$

is generated within the bounding box of the deformed surface. Equation (36) is an empirical formula calibrated so that the number of dipoles per wavelength inside the particle,

$$\mathrm{dpl} = \frac{\lambda_0/\max_c |m_{p,c}|}{d} = \frac{\lambda_{\mathrm{particle,min}}}{d}, \tag{37}$$

where $m_{p,c}$ ($c \in \{x, y, z\}$) is the complex particle refractive index along axis c (cf. Sec. 2.4) and $\lambda_{\mathrm{particle,min}} = \lambda_0/\max_c |m_{p,c}|$ is the shortest wavelength inside the particle, falls roughly in the range 14–20 for ellipsoidal dust particles with the size range detectable by the current CAS-v2 instrument ($r_v \approx 0.1\,\mu\mathrm{m}$ to $0.5\,\mu\mathrm{m}$). For other types of particle and wavelengths, this formula should be customised to ensure adequate dpl. In the current code (v0.3.0), (36) is defined in `shape_model/gaussian_ellipsoid.py`, line 68. Interior points are identified by a three-axis ray-casting algorithm.

7 Mie Theory Reference

For validation, the DDA solution for a volume-equivalent sphere is compared to the exact Mie theory [8].

7.1 Partial-wave expansion

For a homogeneous sphere of radius r_p, the size parameter is $x = k\,r_p = 2\pi\,m_m\,r_p/\lambda_0$ and the relative refractive index is $m_r = m_p/m_m$. The number of partial-wave terms is

$$n_{\max} = \lfloor |x + 4\,x^{1/3} + 2| \rfloor. \tag{38}$$

The Mie coefficients a_n and b_n are computed from Riccati–Bessel functions ψ_n, χ_n, $\xi_n = \psi_n + i\chi_n$ and the logarithmic derivative $D_n(m_r x)$ via downward recurrence [8].

7.2 Efficiencies and scattering amplitudes

$$Q_{\mathrm{ext}} = \frac{2}{x^2}\sum_{n=1}^{n_{\max}} (2n+1)\ \mathrm{Re}(a_n + b_n), \tag{39}$$

$$Q_{\mathrm{sca}} = \frac{2}{x^2}\sum_{n=1}^{n_{\max}} (2n+1)\,(|a_n|^2 + |b_n|^2), \tag{40}$$

$$Q_{\mathrm{abs}} = Q_{\mathrm{ext}} - Q_{\mathrm{sca}}. \tag{41}$$

The angular scattering amplitudes are [8]

$$S_1(\theta) = \sum_{n=1}^{n_{\max}} \frac{2n+1}{n(n+1)}\big(a_n\,\pi_n(\cos\theta) + b_n\,\tau_n(\cos\theta)\big), \tag{42}$$

$$S_2(\theta) = \sum_{n=1}^{n_{\max}} \frac{2n+1}{n(n+1)}\big(a_n\,\tau_n(\cos\theta) + b_n\,\pi_n(\cos\theta)\big), \tag{43}$$

where π_n and τ_n are the angular functions computed via upward recurrence. The forward and backward scattering amplitudes used for validation are obtained from S_1 and S_2 at $\theta = 0$ and $\theta = \pi$.

References

[1] E. M. Purcell and C. R. Pennypacker, "Scattering and absorption of light by nonspherical dielectric grains," *Astrophys. J.* **186**, 705–714 (1973).

[2] B. T. Draine, "The discrete-dipole approximation and its application to interstellar graphite grains," *Astrophys. J.* **333**, 848–872 (1988).

[3] J. J. Goodman, B. T. Draine, and P. J. Flatau, "Application of fast-Fourier-transform techniques to the discrete-dipole approximation," *Opt. Lett.* **16**(15), 1198–1200 (1991).

[4] P. C. Chaumet and A. Rahmani, "Coupled-dipole method for magnetic and negative-refraction materials," *J. Quant. Spectrosc. Radiat. Transfer* **110**, 22–29 (2009).

[5] H. Tadano, T. Sakurai, and Y. Kuramashi, "Block BiCGSTAB(ℓ) for linear systems with multiple right-hand sides," *JSIAM Lett.* **1**, 44–47 (2009).

[6] X.-M. Gu, T.-Z. Huang, L. Li, H.-B. Li, T. Sogabe, and M. Clemens, "Block variants of COCG and COCR methods for solving complex symmetric linear systems with multiple right-hand sides," arXiv:1611.09813 (2016).

[7] K. Muinonen and T. Pieniluoma, "Light scattering by Gaussian random ellipsoid particles: First results with discrete-dipole approximation," *J. Quant. Spectrosc. Radiat. Transfer* **112**, 1747–1752 (2011).

[8] C. F. Bohren and D. R. Huffman, *Absorption and Scattering of Light by Small Particles* (Wiley, New York, 1983).

[9] M. I. Mishchenko, L. D. Travis, and A. A. Lacis, *Scattering, Absorption, and Emission of Light by Small Particles* (Cambridge University Press, 2002).

[10] N. Moteki and K. Adachi, "Measuring the polarized complex forward-scattering amplitudes of single particles in unbounded fluid flow: CAS-v2 protocol," *Opt. Express* **32**(21), 36500–36522 (2024). `https://doi.org/10.1364/OE.533776`.

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